Henrique Baptista

AI Engineer | Agents · LLM Evaluation · Voice AI

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PROFESSIONAL SUMMARY

AI Engineer who builds production agent, LLM, and Voice AI systems, plus the evaluation layer that proves they work. Owning AI problems end to end since 2023, across healthcare and SaaS: LLM-as-Judge evaluators in production, multi-agent systems on LangGraph, ASR evaluation for clinical speech, and live products I design, ship, and operate myself.

PROFESSIONAL EXPERIENCE

ML / AI Engineer - VOA Health — Feb 2025 – Jul 2026

São Paulo, Brazil · HealthTech startup building AI for clinical documentation

The product generated clinical documents at scale, but quality was invisible: failures surfaced only when clinicians noticed. My role was to make quality measurable.

LLM Quality & Evaluation

- Shipped **LLM-as-Judge evaluators to production** (LangSmith online evaluators), scoring AI-generated clinical documents on faithfulness, correctness, and format adherence, so quality regressions were caught before clinicians ever saw them
- Calibrated those judges against human annotation using **Cohen's κ agreement gates** before rollout, and replaced ad-hoc review with a QA gate combining deterministic detectors with a root-cause bug catalog

Voice AI & ASR

- **Owned ASR quality for the product**: benchmarked commercial providers on our own clinical audio and ran the experiments behind each decision, comparing models, VAD configurations, and internal prompts on **WER/CER, latency, and cost**. Generic WER hides what actually matters in a clinical note, so evaluation centered on medication names, where a single wrong word changes the meaning
- Characterized the trade-off that defines streaming clinical ASR: **current audio architectures transcribe markedly better with longer, clearer context**, which is exactly what a live-as-possible pipeline takes away. Tuned VAD and segmentation toward the middle ground where the accuracy gain holds and the user still sees their words appear while speaking

Data Curation & Observability

- Fine-tuning Whisper for clinical audio needed thousands of validated audio-text pairs, and manual exports could never scale. Built the curation pipeline end to end: audio chunking, Hugging Face Hub datasets, a custom **Argilla schema with dual-annotator consensus** and adjudication, and a 24/7 webhook service syncing approved annotations straight to the Hub
- Instrumented the realtime pipeline with Prometheus and Grafana so provider and endpoint choices were made on data instead of guesswork

PROJECTS

WordinAI (solo-built, live at wordinai.com) — LangGraph · deepagents · Next.js · Supabase

- SaaS that turns long-form audio into structured documents, built on a **multi-agent LangGraph system**: a context analyst, a source verifier that requires a reference for every claim, and an action strategist, working over hybrid retrieval (vector plus full-text) of the user's own material
- Operated end to end: multi-provider STT routing with retries and circuit breakers, in-browser Whisper WASM as a zero-cost privacy mode, subscription billing, multi-tenant Supabase RLS, test suites, and Sentry plus PostHog observability

Wilmec (shop-management system for an auto mechanic shop, live in production) — OpenAI Agents SDK · Next.js · Supabase · Whisper

- **Sole engineer** of the system a real repair shop runs on every day, replacing a legacy database and a lot of paper: work orders, vehicle inspection, inventory, and financials, with role-based access and an offline-first architecture, because the users are mechanics on a shop floor with no signal
- Built the AI layer on the **OpenAI Agents SDK**: a conversational inventory agent with Zod-typed tools serving **Telegram, WhatsApp, and web through one engine**, with tool-level guardrails that turn anomalous stock movements and likely duplicates into a confirmation request instead of an action, plus sender allowlist, rate limiting, and an audit trail from every stock movement back to the message that caused it
- Shipped **voice to structured data** (Whisper → strict-JSON LLM → schema-validated write) so mechanics dictate work-order line items from the floor, plus AI diagnostics assembled server-side from each vehicle's service and inspection history, with explicit anti-hallucination rules and every AI interaction logged for later evaluation

Earlier ML & Computer Vision — XGBoost · FastAPI · Detectron2 · AWS

- An education-risk early-warning system for an NGO (XGBoost, **92.59% accuracy / 0.97 ROC-AUC**, served over FastAPI with drift monitoring on Hugging Face Spaces), and an insurance-claim segmentation pipeline built on Detectron2 with a COCO dataset made from scratch, cutting manual review by roughly 80% (Porto Seguro Challenge, **Top-10 finalist**)

EDUCATION

Postgraduate Program, Machine Learning Engineering - FIAP — Completed Mar 2026

Bachelor, Systems Analysis & Development - FIAP — Completed Dec 2024

TECHNICAL SKILLS

LLM & Agents:	LangGraph, LangChain, deepagents, OpenAI Agents SDK, OpenAI / Anthropic / Groq APIs, Model Context Protocol (MCP), tool calling, structured output, multi-agent orchestration, prompt engineering & caching
RAG & Evaluation:	Retrieval-Augmented Generation (RAG), embeddings, chunking strategies, hybrid (vector + full-text) search, ChromaDB, pgvector, Pinecone; LLM-as-Judge, LangSmith, evaluation datasets & golden sets, WER/CER (jiwer), human-annotation consensus (Argilla), Cohen's κ calibration, ablation & failure analysis
ML, Speech & Vision:	PyTorch, Hugging Face (Transformers, Datasets, Hub), scikit-learn, XGBoost, pandas, NumPy, MLflow, feature engineering, cross-validation, class imbalance; Whisper / WhisperX, VAD (Silero), diarization (pyannote), FFmpeg audio pipelines, fine-tuning data curation; Detectron2, Mask R-CNN, YOLO, COCO
MLOps & Infra:	Docker, FastAPI, CI/CD, Git, AWS (S3, EC2), Railway, Vercel, Hugging Face Spaces, Prometheus / Grafana, Sentry, PostHog, drift monitoring (Evidently), Streamlit / Gradio, Jupyter
Engineering:	Python, TypeScript, Next.js, PostgreSQL, Supabase, REST APIs, Zod / Pydantic validation, row-level security